

Homogeneity Fusion for Sparse Grouped Network VAR models

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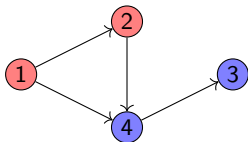
- Directed network with adjacency matrix $A = (a_{ij})_{1 \leq i, j \leq d}$.
- Latent partition $\mathcal{G} = \{G_1, \dots, G_K\}$ of $\{1, \dots, d\} =: [d]$ into K groups. We denote the group membership of node i by g_i .
- GNAR model [Zhu et al., 2023]:

$$Y_{i,t} = \sum_{j=1}^d \phi_{g_i g_j} \frac{a_{ij}}{n_i} Y_{j,t-1} + \nu_{g_i} Y_{i,t-1} + \varepsilon_{i,t}, \quad i \in [d], \quad t \in [T],$$

where $n_i = \sum_{j=1}^d a_{ij}$ the out-degree of i , and $\varepsilon_{i,t} \stackrel{\text{i.i.d.}}{\sim} N(0, \sigma^2)$.

- Stationarity condition: $\max_{k \in [K]} (\max_{l \in [K]} |\phi_{kl}| + |\nu_k|) < 1$

Example



Updating equations:

$$\begin{cases} Y_{1,t} = \frac{\phi_{11}}{2} Y_{2,t-1} + \frac{\phi_{12}}{2} Y_{4,t-1} + \nu_1 Y_{1,t-1} + \varepsilon_{1,t} \\ Y_{2,t} = \phi_{12} Y_{4,t-1} + \nu_1 Y_{2,t-1} + \varepsilon_{2,t} \\ Y_{3,t} = \nu_2 Y_{3,t-1} + \varepsilon_{3,t} \\ Y_{4,t} = \phi_{22} Y_{3,t-1} + \nu_2 Y_{4,t-1} + \varepsilon_{4,t} \end{cases}$$

Contributions

- Previous work assumed known network. Here we estimate the groups $\mathcal{G}$, parameters (Φ, ν) , and network A on from time series data, via the lens of **homogeneity fusion**.

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Contributions

- Previous work assumed known network. Here we estimate the groups $\mathcal{G}$, parameters (Φ, ν) , and network A on from time series data, via the lens of **homogeneity fusion**.
- **Theoretical** guarantees for fusion estimator (significantly more challenging than in previous works, due to VAR structure).
- **Implementation**: the use of Mixed Integer Programming (MIP) improves the estimator's performance compared to the iterative algorithm of [Zhu et al., 2023].

Estimation

Simultaneous estimation by solving the following optimization problem.

Fusion estimator

$$\begin{aligned} & \underset{\substack{A \in \{0,1\}^{d \times d}, \mathcal{G} \in [K]^d, \\ \Phi \in \mathbb{R}^{K \times K}, \nu \in \mathbb{R}^K}}{\operatorname{argmin}} \sum_{t=1}^T \sum_{i=1}^d \left(Y_{i,t} - \sum_{j=1}^d \phi_{g_i g_j} \frac{a_{ij}}{n_i} Y_{j,t-1} - \nu_{g_i} Y_{i,t-1} \right)^2 \\ & \text{s.t.} \quad \sum_{i=1}^d \sum_{j=1}^d a_{ij} \leq s \end{aligned}$$

Goals

- **Theory:** The *misspecification probability* for grouping/network vanishes:

$$\mathbb{P} \left((\hat{\mathcal{G}}, \hat{A}) \neq (\mathcal{G}^*, A^*) \right) \rightarrow 0, \quad \text{as } d, T \rightarrow \infty,$$

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- **Implementation:** Solve the fusion optimization problem at scale by getting an initial estimate of the network via Sure Independence Screening (SIS) or Best Subset Selection (BSS), estimating the group memberships and parameters via MIP, and iterating these steps.

Regression forms

- The GNAR model can equivalently be written as

$$Y = Z\beta + \varepsilon,$$

where

$$\beta = (b_{11}, \dots, b_{1d}, \dots, b_{d1}, \dots, b_{dd})', \quad b_{ij} = \begin{cases} \phi_{g_i g_j} \frac{a_{ij}}{n_i}, & \text{if } i \neq j, \\ \nu_{g_i}, & \text{if } i = j \end{cases}$$

$$Y = \text{vec} \begin{pmatrix} Y'_1 \\ \vdots \\ Y'_T \end{pmatrix}, \quad X = \begin{pmatrix} Y'_0 \\ \vdots \\ Y'_{T-1} \end{pmatrix}, \quad Z = I_d \otimes X, \quad \varepsilon = \text{vec} \begin{pmatrix} \varepsilon'_1 \\ \vdots \\ \varepsilon'_T \end{pmatrix}$$

- Reduced form:

$$Y = Z_{\mathbb{G}}\beta_{\mathbb{G}} + \varepsilon,$$

where $Z_{\mathbb{G}}$ and $\beta_{\mathbb{G}}$ are of reduced dimension according to the network-grouping parameter $\mathbb{G} = (\mathcal{G}, A)$.

Theory

1. Bound the misspecification probability:

$$\mathbb{P} \left((\hat{\mathcal{G}}, \hat{A}) \neq (\mathcal{G}^*, A^*) \right) \leq \sum_{\substack{(\mathcal{G}, A) \neq (\mathcal{G}^*, A^*) \\ |\mathcal{G}| = |\mathcal{G}^*|, \|A\|_0 \leq \|A^*\|_0}} \mathbb{P} \left(\min_{\tilde{\beta}: (\tilde{\mathcal{G}}, \tilde{A}) = (\mathcal{G}, A)} \|Y - Z\tilde{\beta}\|_2^2 < \|Y - Z\hat{\beta}_{\text{ol}}\|_2^2 \right).$$

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2. Rewrite probability as

$$\begin{aligned} \mathbb{P} \left(\min_{\tilde{\beta}: (\tilde{\mathcal{G}}, \tilde{A}) = (\mathcal{G}, A)} \|Y - Z\tilde{\beta}\|_2^2 < \|Y - Z\hat{\beta}_{\text{ol}}\|_2^2 \right) &= \mathbb{P} \left(\|Y - Z\hat{\beta}\|_2^2 < \|Y - Z\hat{\beta}_{\text{ol}}\|_2^2 \right) \\ &= \mathbb{P} \left(\varepsilon' (P_{\mathbb{G}} - P_{\mathbb{G}^*}) \varepsilon - 2\varepsilon' (I - P_{\mathbb{G}}) Z\beta^* > \|(I - P_{\mathbb{G}}) Z\beta^*\|_2^2 \right), \end{aligned}$$

where $P_{\mathbb{G}} := Z_{\mathbb{G}}(Z'_{\mathbb{G}}Z_{\mathbb{G}})^{-1}Z'_{\mathbb{G}}$ is a $dT \times dT$ projection matrix including all lagged values.

Remarks

1. In past work, products of the form $(Z'_G Z_G)^{-1} Z'_G Z_{G^*}$ were deterministic. Now they are random and have a non-trivial dependence structure, so we replace them by $(\mathbb{E}[Z'_G Z_G])^{-1} \mathbb{E}[Z'_G Z_{G^*}]$ and use concentration inequalities to control the difference.

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2. The main technical difficulty lies in the dependence structure between ε and Z . To handle this, we carefully express Z as function of ε and then exploit the Hanson-Wright inequality.
3. We handle the complicated matrices Z_G by decomposing them into $Z_G = Z N_G$, where N_G is a deterministic matrix reflecting the network-grouping structure. Results involving Z can be handled through [Basu and Michailidis, 2015], and we develop technical results to capture the behavior of N_G .

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3. Iterate the previous steps.

MIP formulation (known network)

$$\begin{aligned}
 & \underset{\substack{B \in \mathbb{R}^{d \times d}, \Phi \in \mathbb{R}^{K \times K}, \\ \nu \in \mathbb{R}^K, Z \in \{0,1\}^{d \times K}}}{\operatorname{argmin}} && \sum_{t=1}^T \sum_{i=1}^d \left(Y_{i,t} - \sum_{j=1}^d b_{ij} Y_{j,t-1} \right)^2 \\
 & \text{s.t.} && z_{ik} \in \{0,1\}, \quad i \in [d], k \in [K] \\
 & && \sum_{k=1}^K z_{ik} = 1, \quad i \in [d] \\
 & && b_{ii} - \nu_k \leq M(1 - z_{ik}), \quad i \in [d], k \in [K] \\
 & && b_{ii} - \nu_k \geq -M(1 - z_{ik}), \quad i \in [d], k \in [K] \\
 & && n_i b_{ij} - \phi_{kl} a_{ij} \leq M(2 - z_{ik} - z_{jl}), \quad i, j \in [d], i \neq j, k, l \in [K] \\
 & && n_i b_{ij} - \phi_{kl} a_{ij} \geq -M(2 - z_{ik} - z_{jl}), \quad i, j \in [d], i \neq j, k, l \in [K]
 \end{aligned}$$

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- Network estimation: SIS or BSS
- Parameters/Groups estimation: MIP or iterative method of [Zhu et al., 2023].
- Our MIP estimator achieves low misclassification and estimation errors, even in high-dimensional-low-sample-size regimes.
- MIP outperforms the method of [Zhu et al., 2023] for various sparsity levels and sample sizes.

Conclusions

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- Our key contribution is a **joint estimation framework**, combining group structure, network topology, and autoregressive dynamics, with strong theoretical guarantees on misspecification error.
- We employed **mixed integer programming** to solve our estimation problem at scale.
- MIP outperforms existing methods in simulations and reveals interesting structures in real financial & macroeconomic datasets.



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